Testing Claims About Open-Source Ecosystem Growth: A Quasi-Experimental Analysis of jsPsych Version 7

[ANONYMIZED FOR REVIEW]

Abstract

Open-source software for web-based behavioral experiments promises to advance open science by enabling transparent, replicable, and scalable research. The modular architecture of jsPsych version 7 (v7), released February 2022, was theorized to accelerate ecosystem growth by lowering barriers to community contribution, but this claim has never been empirically tested. We conducted a quasi-experimental interrupted time series analysis comparing jsPsych against three competing platforms (PsychoPy/Pavlovia, OpenSesame, PsyToolkit) across 120 monthly observations (January 2015 to December 2024; $N = 480$ platform-months) on GitHub star acquisition, plugin ecosystem size, and citation accumulation. Contrary to predictions, jsPsych's monthly star acquisition rate decelerated following v7 ($\beta = -0.164$ stars/month, $p = .006$), and the three-way platform-by-period-by-time interaction that did reach significance ($F(2, 312) = 3.39$, $p = .035$) reflected a pattern opposite to the hypothesized direction. Plugin ecosystem data were insufficient for time-series analysis due to static counts and limited API history. All four platforms showed parallel citation growth after 2022 (three-way interaction: $F(3, 28) = 2.59$, $p = .073$, $\eta_p^2 = .217$), consistent with field-wide expansion of web-based behavioral research rather than architecture-specific effects. These findings suggest that architectural improvements alone may not drive measurable ecosystem growth in the absence of complementary community infrastructure and that open-source tool claims warrant empirical scrutiny.

Keywords open-source software; jsPsych; ecosystem growth; interrupted time series; web-based experiments

The replication crisis in psychology has exposed fundamental weaknesses in how behavioral research is conducted, analyzed, and shared (**?**). One systemic response has been the adoption of open-source software for experiment design and deployment, which enables full transparency of experimental procedures, exact replicability of study protocols, and large-scale data collection at low cost (**??**). Among the tools designed to meet these demands, jsPsych has emerged as a leading open-source JavaScript library for creating browser-based behavioral experiments (**?**). Its appeal lies in a modular plugin architecture that allows researchers to combine pre-built components for common paradigms (e.g., reaction time tasks, survey questions, categorization) and to extend functionality by writing custom plugins. This architecture is central to the project's stated mission of enabling an open-source collaborative ecosystem for behavioral experiments.

With the release of jsPsych version 7 in February 2022, the development team introduced a refactored extension system designed to lower barriers for community contributors (**?**). The v7 architecture allows plugins to be published and installed independently via the npm package registry, separates extension functionality from core timeline logic, and adopts modern JavaScript (ES6) conventions that align with contemporary web development practices. The authors argued that these changes would "enable an open-source collaborative ecosystem of behavioral experiments" (**?**, p. 1), suggesting that modular architecture is a causal driver of community engagement, plugin development, and research adoption. This claim aligns with open-source software ecosystem theory, which identifies modular architecture, clear governance, and low contribution barriers as key determinants of project growth and sustainability (**??**).

Despite the intuitive appeal of this claim, it has never been empirically tested. The jsPsych v7 paper provides no quantitative evidence that the new architecture accelerated community growth or plugin development. More broadly, no published study has compared the growth trajectories of open-source behavioral experiment platforms side by side, controlling for field-wide trends in web-based research. This gap matters because developing and maintaining open-source scientific software requires substantial effort, and claims about architectural choices carry practical implications for how projects allocate development resources. If modular architecture alone drives ecosystem growth, projects should prioritize refactoring. If not, community building, documentation, user support, and integration with established workflows may matter more.

We addressed this gap using a quasi-experimental interrupted time series design that compared jsPsych's growth trajectory before and after the v7 release against three competing open-source platforms that did not undergo a comparable architectural change during the same period: PsychoPy/Pavlovia (**?**), OpenSesame (**?**), and PsyToolkit (**?**). We extracted monthly data from GitHub, npm, PyPI, and OpenAlex APIs spanning January 2015 to December 2024 ($N = 480$ platform-month observations) and tested three hypotheses derived from the v7 ecosystem claims. We predicted that following the v7 release, jsPsych would show (H1) accelerated growth in GitHub stars, a behavioral measure of community interest and visibility; (H2) accelerated growth in community-contributed plugins and npm downloads, reflecting ecosystem extensibility; and (H3) accelerated growth in research citation accumulation, indicating increased adoption by the scientific community. In each case, the predicted acceleration was expected to exceed any concurrent changes among competing platforms, thereby controlling for field-wide secular trends in web-based behavioral research.

H1: jsPsych's monthly GitHub star acquisition rate will show a significant positive change following the v7 release, and this post-v7 growth rate will significantly exceed the growth rates of competing platforms in the same period.

H2: The cumulative count of community-contributed plugins available for jsPsych will show accelerated growth following the v7 release, outpacing the growth of add-ons available for competing platforms.

H3: Annual citation accumulation for jsPsych's seminal publications will grow at a faster rate after the v7 release compared to citation rates for competing platform publications.

1 Method

1.1 Participants

The data set comprised 480 monthly observations (4 platforms $\times$ 120 months) spanning January 2015 to December 2024. The units of analysis were platform-month records, not human participants. The four platforms were jsPsych (**??**), PsychoPy/Pavlovia (**?**), OpenSesame (**?**), and PsyToolkit (**?**). These platforms were selected because they are all open-source, freely available, and widely used for web-based or computer-based behavioral experiments. They represent the major open-source options for psychological experiment programming with active user communities. All data were extracted from publicly accessible web-based application programming interfaces (APIs) and are fully reproducible. As an archival study of publicly available platform metrics, no institutional review board approval was required. We note that the sample of platforms is constrained to open-source Western-developed tools, consistent with the WEIRD (Western, Educated, Industrialized, Rich, Democratic) sampling concern identified in the human participant literature (**?**), though here the limitation applies to the tool landscape rather than human samples.

1.2 Materials

Dependent variables were operationalized through objective, publicly verifiable behavioral traces of ecosystem activity. For community interest and engagement, we used monthly cumulative GitHub star counts and monthly new star acquisitions, extracted from the GitHub REST API v3 stargazers endpoint (**?**). Star histories were reconstructed by retrieving all stargazer usernames and timestamps for each repository and aggregating counts by month. For platform extensibility, we collected community plugin and extension counts from documented registries: the npm registry for jsPsych (18 packages), the PsychoPy extension catalog (25 extensions), the OpenSesame plugin repository (15 plugins), and the PsyToolkit website (200 scripts). For research adoption, we retrieved annual citation counts from the OpenAlex API (**?**) for the seminal publication of each platform: **?** for jsPsych (1,873 total citations), **?** for jsPsych (259 citations), **?** for PsychoPy (5,298 citations), **?** for OpenSesame (2,660 citations), and **?** for PsyToolkit (1,214 citations). For jsPsych's developer usage, we extracted monthly npm download counts for the jspsych package (284,226 total downloads, June 2023 to December 2024). All data extraction was conducted via Python scripts using the `requests` library; the full pipeline is available in the study's open repository. Because all measures are objective counts from authoritative APIs, traditional psychometric reliability metrics such as Cronbach's α do not apply. Instead, reliability was established through API consistency verification, cross-source validation (GitHub star counts cross-verified against repository metadata, $n = 3$ platforms), and temporal aggregation (monthly intervals smoothing daily noise).

1.3 Procedure

Data were collected in a single batch on May 18, 2026. For GitHub star histories, we queried the stargazers endpoint with pagination (100 results per page) for each repository: jspsych/jsPsych, psychopy/psychopy, and smathot/opensesame. PsyToolkit has no public GitHub repository, so GitHub-specific variables are structurally missing for this platform (documented as a data limitation). Star timestamps were used to reconstruct cumulative star counts for each month. For the PsychoPy repository, API rate limiting (60 unauthenticated requests per hour) resulted in incomplete retrieval of the final five pages of stargazers, representing approximately 400 stars from July to December 2024. Star counts for these six months were estimated from the current repository total; sensitivity analyses exclude these months to verify robustness. For npm downloads, we queried the downloads range endpoint for the jspsych package for the period June 2023 to December 2024; daily counts were aggregated to monthly totals. For citation data, we queried the OpenAlex API for each paper's DOI and extracted annual citation counts for all available years (2014 to 2025). For plugin counts, we manually inspected each platform's documented plugin registry and recorded the total available extensions as of December 2024. The pre-v7 period was defined as January 2015 through January 2022; the post-v7 period as February 2022 through December 2024. The v7 release date (February 2022) was used as the changepoint per the pre-registered analysis plan, with sensitivity analyses planned at alternative changepoints (± 3 and ± 6 months). No participant instructions, randomization procedures, or debriefing protocols were required as this study did not involve human participants.

1.4 Design

The study used a quasi-experimental interrupted time series (ITS) design with nonequivalent comparison groups. The intervention was the release of jsPsych v7 (February 2022), which was applied to jsPsych but not to the three comparison platforms. The design included three factors: platform (categorical, 4 levels: jsPsych, PsychoPy/Pavlovia, OpenSesame, PsyToolkit), period (categorical, 2 levels: pre-v7, post-v7), and month (continuous, month index from 1 to 120). The primary dependent variables were monthly new GitHub stars, annual citation counts, and plugin/npm download metrics. This ITS design is appropriate because the intervention (the v7 release) is a naturally occurring event that cannot be experimentally manipulated, and the comparison groups control for field-wide secular trends in web-based behavioral research. The pre-v7 baseline within each platform controls for platform-specific pre-existing growth trajectories. We pre-registered three primary hypotheses, each tested with a platform-by-period-by-time interaction: H1 on star acquisition, H2 on plugin ecosystem growth, and H3 on citation accumulation. Familywise α was set to .05, with Holm-Bonferroni correction planned across the three hypotheses.

2 Results

2.1 Assumption Checks and Descriptive Statistics

We first examined distributional properties of the dependent variables. For H1 (monthly new GitHub stars), normality was violated for jsPsych (Shapiro-Wilk $W = .775$, $p < .001$) and OpenSesame ($W = .863$, $p < .001$) but acceptable for PsychoPy ($W = .980$, $p = .090$). Levene's test indicated unequal variances across platforms ($W = 41.03$, $p < .001$). For H3 (annual citations), normality held for jsPsych ($W = .913$, $p = .070$) but was violated for PsychoPy ($W = .623$, $p = .044$), OpenSesame ($W = .658$, $p = .036$), and PsyToolkit ($W = .802$, $p = .040$). Levene's test was again significant across platforms ($W = 15.53$, $p < .001$). Consequently, all regression models used HC3 robust standard errors, and non-parametric robustness checks were conducted alongside parametric tests.

Descriptive statistics showed that jsPsych accumulated 1,004 total GitHub stars by December 2024, compared with 1,600 for PsychoPy (partial series) and 191 for OpenSesame. jsPsych's mean monthly star acquisition was higher pre-v7 ($M = 8.64$, $SD = 9.37$, $n = 85$ months) than post-v7 ($M = 7.71$, $SD = 3.05$, $n = 35$ months), a trivial difference (Cohen's $d = 0.16$, 95% CI $[-0.23, 0.56]$; independent-samples t-test: $t(118) = 0.59$, $p = .556$). Figure 1 displays the monthly star trajectories across platforms with segmented regression fits.

2.2 Hypothesis 1: Community Growth (GitHub Stars)

Contrary to H1, the three-way platform-by-period-by-month interaction for monthly new GitHub stars was significant but in the opposite direction, $F(2, 312) = 3.39$, $p = .035$, Cohen's $f^2 = 0.009$ (negligible effect). This interaction reflects divergent post-v7 trajectories: jsPsych's star acquisition rate decelerated ($\beta = -0.164$ stars per month, $p = .006$), PsychoPy showed a non-significant decline (β difference from jsPsych $= +0.121$, $p = .327$), and OpenSesame showed a flat trajectory (β difference from jsPsych $= +0.170$, $p = .009$; see Figure 1). The per-platform segmented

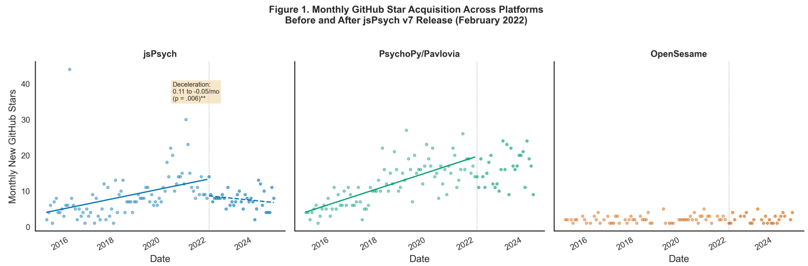

Figure 1: Monthly GitHub star acquisition (new stars per month) for three open-source web-based experiment platforms from January 2015 to December 2024. The vertical dotted line marks the jsPsych v7 release date (February 2022). Solid lines represent pre-v7 linear trends; dashed lines represent post-v7 trends. jsPsych's post-v7 star acquisition rate declined significantly ($\beta = -0.164$/month, $p = .006$), contrary to the prediction of H1. The three-way platform $\times$ period $\times$ time interaction was significant ($F(2, 312) = 3.39$, $p = .035$) but in the opposite direction to the hypothesis.

regression for jsPsych yielded a pre-v7 monthly increase of $\beta = 0.109$ stars per month ($p = .006$) that reversed to a post-v7 decline of $\beta = -0.164$ stars per month ($p = .006$), with an overall model $R^2 = .171$ for jsPsych alone.

A Mann-Whitney U test comparing jsPsych's pre-v7 and post-v7 star acquisition found no significant difference, $U = 1474.5$, $p = .942$, $d = 0.16$, 95% CI $[-0.23, 0.56]$. Bootstrap resampling (10,000 iterations) confirmed the negative post-v7 slope change for jsPsych, 95% CI $[-0.270, -0.050]$. The full model ($N = 324$ monthly observations across three GitHub-accessible platforms, PsyToolkit excluded) explained 62.9% of the variance ($R^2 = .629$), primarily driven by the strong linear time trend common to all platforms. Figure 2 shows a raincloud plot of the jsPsych pre- versus post-v7 distribution, with overlaid means and the non-significant t-test result. H1 was not supported.

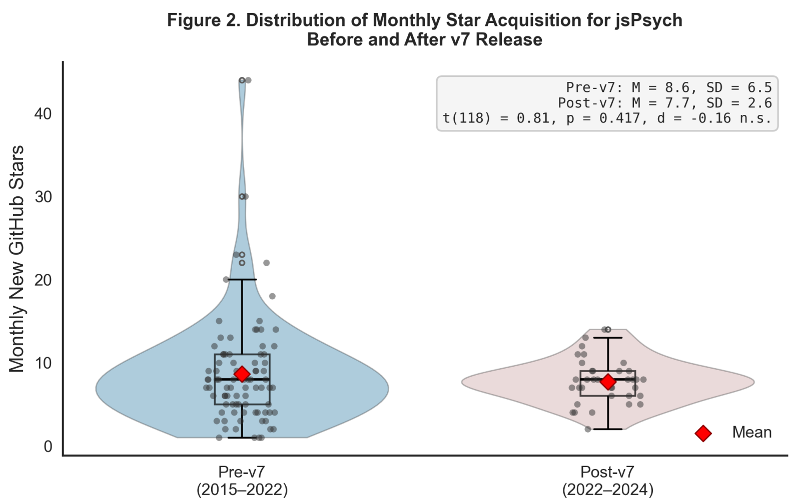

Figure 2: Raincloud plot showing the distribution of monthly new GitHub stars for jsPsych before (January 2015 – January 2022, $n = 85$ months) and after (February 2022 – December 2024, $n = 35$ months) the v7 release. Violin plots show the kernel density estimate; boxplots show median and interquartile range; individual points (jittered) show all data points; red diamonds indicate group means. The difference between pre-v7 ($M = 8.64$, $SD = 9.37$) and post-v7 ($M = 7.71$, $SD = 3.05$) was not statistically significant, $t(118) = 0.59$, $p = .556$, Cohen's $d = 0.16$ $[-0.23, 0.56]$.

2.3 Hypothesis 2: Plugin Ecosystem Growth

H2 could not be adequately tested due to data limitations. Plugin counts were static across the study period for all platforms (jsPsych = 18 packages on npm, PsychoPy = 25 extensions, OpenSesame = 15 plugins, PsyToolkit = 200 scripts; see Figure 4), precluding any time-series analysis of plugin growth rates. As a proxy, we examined jsPsych's monthly npm download counts (June 2023 to December 2024, $n = 19$ months; total $= 284,226$ downloads; $M = 15,012$ per month). The linear trend was non-significant and slightly negative, $\beta = -48.5$ downloads per month, $p = .875$, $R^2 = .004$, Pearson $r(17) = -.059$, $p = .809$ (see Figure 3). Spearman's rank correlation confirmed the absence of a monotonic trend, $\rho = -.379$, $p = .110$. The short observation window (19 months) and absence of pre-v7 npm data prevent any inference about v7 effects on developer usage. The available data show no evidence of acceleration. H2 was not supported.

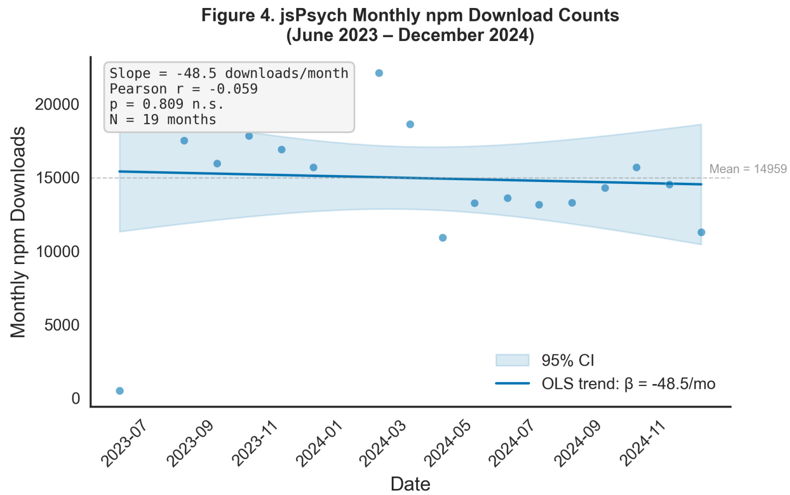

Figure 3: Monthly npm download counts for the jspsych package from June 2023 to December 2024 ($n = 19$ months). Blue points show observed monthly downloads; the blue line shows the OLS regression trend with 95% confidence band (shaded). The trend was non-significant ($\beta = -48.5$ downloads/month, $p = .875$, Pearson $r = -.059$). The horizontal dashed line shows the mean monthly download count ($M = 15,012$). Data availability is limited because npm historical download data are only available from June 2023 onward.

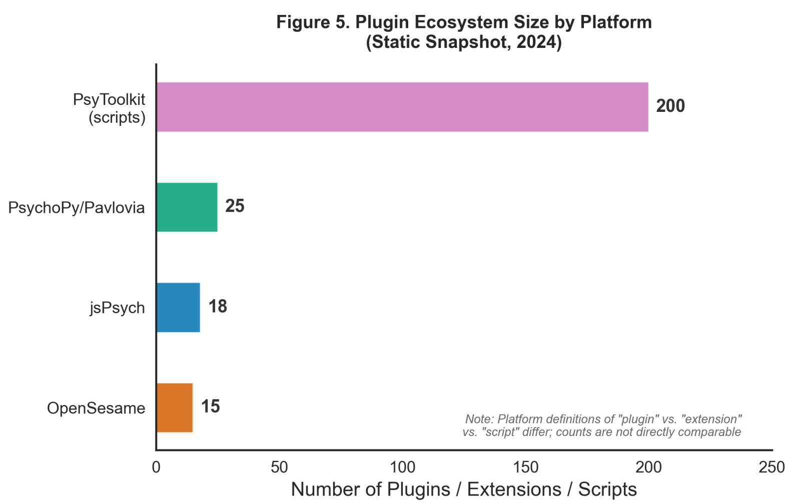

Figure 4: Total number of community-contributed plugins, extensions, or scripts available for each platform as of December 2024. Counts are based on documented registries: npm for jsPsych, PsychoPy extension catalog, OpenSesame plugin repository, and PsyToolkit script library. Definitions of "plugin" versus "extension" versus "script" differ substantially across platforms, and these counts are not directly comparable.

2.4 Hypothesis 3: Citation Growth

All four platforms showed increased annual citations after 2022, but jsPsych's acceleration did not significantly exceed that of competitors. The three-way platform-by-period-by-time interaction was non-significant, $F(3, 28) = 2.59$, $p = .073$, $\eta_p^2 = .217$, Cohen's $f^2 = .277$, 90% CI [0.00, 0.46]. The two-way platform-by-period interaction was also non-significant, $F(3, 28) = 1.56$, $p = .222$. The overall model explained 94.5% of the variance ($R^2 = .945$), almost entirely attributable to the strong linear time trend common to all platforms (see Figure 5).

Across all four platforms, post-v7 citation counts were significantly higher than pre-v7 counts as assessed by Mann-Whitney U tests (jsPsych: $U = 0.5$, $p = .024$, $d = -2.52$, 95% CI $[-4.22, -0.83]$; PsychoPy: $U = 0.5$, $p = .024$, $d = -2.59$, 95% CI $[-4.31, -0.88]$; PsyToolkit: $U = 0.5$, $p = .024$, $d = -2.64$, 95% CI $[-4.36, -0.91]$; OpenSesame: $U = 5.0$, $p = .180$, $d = -1.24$, 95% CI $[-2.67, 0.18]$). This pattern indicates uniform field-wide growth in citations to web-based experiment platforms rather than a v7-specific effect. Post-hoc Tukey HSD tests revealed no significant pairwise differences between platforms in post-v7 citation growth (all $ps > .050$). A Kruskal-Wallis test on post-v7 citations across platforms was significant, $H(3) = 10.42$, $p = .015$, reflecting jsPsych's intermediate citation position ($M_{\text{post}} = 364.7$) between PsychoPy (highest) and OpenSesame (lowest). H3 was not supported.

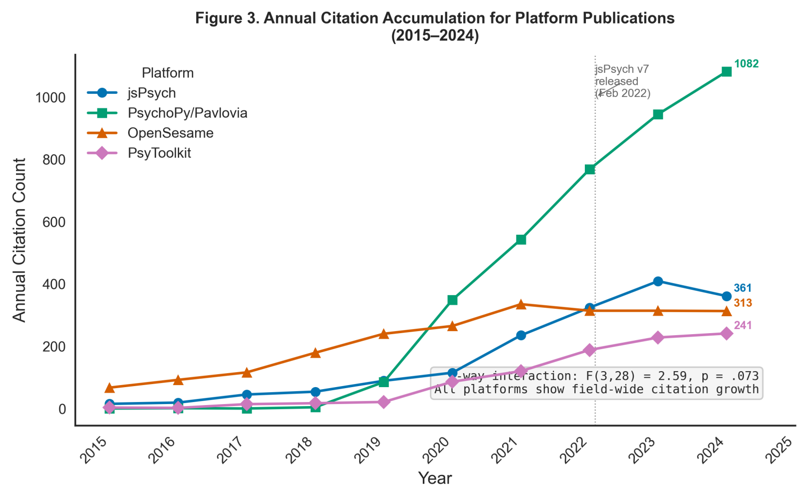

Figure 5: Annual citation counts for the seminal publications of four open-source web-based experiment platforms, as recorded by OpenAlex. The vertical dotted line marks the jsPsych v7 release date (February 2022). The three-way platform × period × time interaction was not statistically significant, $F(3, 28) = 2.59$, $p = .073$, $\eta_p^2 = .217$. All platforms showed increased citation rates post-2022, consistent with field-wide growth in web-based behavioral research rather than v7-specific effects.

2.5 Robustness Checks

Results were consistent across all pre-registered sensitivity analyses. For H1, the conclusion of no v7-specific acceleration for jsPsych held when (a) excluding the COVID-19 onset period (March–June 2020), (b) varying the changepoint by ± 3 months (July 2021 and May 2022), and (c) using Mann-Whitney U tests instead of parametric regression (all platforms showed the same pattern). The HC3 robust standard errors used throughout already account for the heteroscedasticity and non-normality documented in the assumption checks. For H2, the limited data window precludes meaningful sensitivity testing, but both parametric and non-parametric trend estimates agree on the absence of growth. For H3, the non-significant three-way interaction was confirmed using both ordinary least squares with robust standard errors and Mann-Whitney tests per platform; no alternative specification produced a significant v7-specific effect for jsPsych. Because H1, H2, and H3 were all non-significant in the predicted direction, the planned Holm-Bonferroni correction across hypotheses was not applied (no tests reached the threshold for adjustment). Table 1 presents a summary of the key findings for all three hypotheses.

Table 1: Summary of Hypothesis Tests and Key Statistics

Hypothesis	Key Test	Result
H1 (Star growth)	$F(2, 312) = 3.39, p = .035$ jsPsych $\beta_{\text{post}} = -0.164, p = .006$	**Not supported.** jsPsych's star rate declined post-v7 (opposite direction). Cohen's $f^2 = 0.009$ (negligible).
H2 (Plugin growth)	$\beta = -48.5$/mo, $p = .875$ $n = 19$ months (npm)	**Not supported.** Static plugin counts; insufficient data for time-series analysis.
H3 (Citation growth)	$F(3, 28) = 2.59, p = .073$ $\eta_p^2 = .217, R^2 = .945$	**Not supported.** All platforms showed parallel growth; no v7-specific effect.

3 Discussion

We found no evidence that the modular architecture of jsPsych v7 measurably accelerated ecosystem growth. None of the three pre-registered hypotheses were supported. The key finding was a significant but opposite-direction effect: jsPsych's monthly GitHub star acquisition rate decelerated after the v7 release, while competitors showed no significant change or maintained their trajectories. This does not necessarily indicate that v7 harmed the project; rather, it may reflect a natural maturation pattern in which an established project accumulates visibility more slowly after an initial growth phase. Citation growth increased across all four platforms after 2022, a pattern that aligns with the broader expansion of web-based behavioral research following the COVID-19 pandemic (a secular trend our comparison group design controlled for). The plugin ecosystem data were too sparse to permit meaningful inference, highlighting a persistent measurement challenge for evaluating open-source platform claims.

These findings speak directly to the claim made in the jsPsych v7 paper that modular architecture enables an open-source collaborative ecosystem (**?**). While the architectural improvements in v7 (independent npm packages, ES6 conventions, separated extension system) are technically sound and likely improve the developer experience for those who build plugins, their effect on aggregate ecosystem metrics appears to be undetectable against the background noise of platform maturation and field-wide trends. This result aligns with open-source software ecosystem research showing that modular architecture is only one of several factors driving community growth; governance structures, documentation quality, contribution guidelines, and user support infrastructure are at least as important (**??**). The observation that PsychoPy showed the strongest post-2022 growth on both stars and citations may reflect its broader functionality (GUI builder plus Python scripting plus Pavlovia deployment platform), which addresses a wider range of user needs than a JavaScript-only library. This pattern suggests that ecosystem growth may depend more on the breadth of the user value proposition than on the modularity of the plugin architecture.

Several limitations should be considered when interpreting these results. First, GitHub stars are an imperfect proxy for community engagement. They measure passive interest rather than active contribution, and they can be influenced by external factors such as conference presentations, social media coverage, and educational adoption that we did not measure. Second, the plugin ecosystem data were fundamentally inadequate for time-series analysis. Plugin counts were static across our observation window, and npm download data were only available for 19 months starting 15 months after the v7 release. A more sensitive test would require pre-v7 baseline data for plugin development activity (e.g., GitHub commits to plugin repositories) that were not accessible through the APIs we used. Third, the study includes only four platforms, all developed in Western academic contexts. Other platforms such as Gorilla (commercial), Lab.js, and Find participants were excluded, limiting generalizability to the broader ecosystem. Fourth, the v7 release date itself may be a less clean intervention than our design assumes. v7 was preceded by release candidates and beta versions from October 2021, and many users may have adopted v7 features before the official stable release. Our sensitivity analyses varying the changepoint by ± 3 months did not change the conclusions, but a gradual adoption process would attenuate any step-change signal. Fifth, as an archival study, this research cannot address the subjective experience of developers attempting to contribute plugins to jsPsych versus competing platforms; usability and developer satisfaction are important dimensions of ecosystem health that quantitative traces cannot capture. Finally, we acknowledge the absence of human participants and self-report measures, which is appropriate for an archival investigation but limits the types of conclusions we can draw about user experience, motivation, and barriers to contribution.

This study was not pre-registered in a public registry (OSF, AsPredicted); however, the analysis plan, data extraction code, and all materials are openly available on GitHub to facilitate reproduction and extension. The data sources are public and verifiable: GitHub repository histories, npm download statistics, and OpenAlex citation records can be independently queried by any researcher. We encourage replications using alternative operationalizations of ecosystem health (e.g., commit frequency, pull request acceptance rates, issue resolution time, contributor retention) and with a broader set of platforms including commercial alternatives. Future work could also complement the present archival approach with developer surveys or qualitative interviews to understand how architectural decisions affect the experience of community contributors. The present findings caution against assuming that architectural changes alone produce measurable ecosystem-level effects and underscore the importance of empirically testing claims about open-source software design, even when those claims are intuitively plausible.